DEPI GRADUATION PROJECT

Task 3 - Requirements Gathering

Automated Teacher Workflow with n8n

DEPI Recommended Capstone Project #9 - AI in Education

Track	AI in Education
Team Size	5 Members
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Group Code	GIZ5_IND1_G1
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1. Requirements Gathering Overview

This document defines who the project serves, how users will interact with the workflow, and what the system must do. It is based on the approved project scope from Task 1: a teacher submits instructional information and an n8n workflow generates a lesson outline, student activities, an assessment, an answer key, and a differentiated version, then organizes the outputs for teacher review.

Core Requirement Boundary
The core project is an n8n automation, not a full LMS or student-facing platform. External API or educational-platform integration remains optional until the core workflow is stable.

2. Stakeholder Analysis

Stakeholder	Role	Interaction	Key Needs	Influence	Interest
Teacher / Teaching Professional	Primary user	Enter lesson information; receive and review generated materials.	Fast preparation, clear outputs, relevant content, control over inputs, understandable errors.	High	High
Students	Indirect beneficiary	Benefit from activities, assessments, and differentiated materials after teacher review.	Age/level-appropriate materials, clarity, appropriate differentiation.	Medium	Medium
Technical Instructor / Project Reviewer	Evaluator / advisor	Review scope, requirements, implementation evidence, and project quality.	Clear documentation, traceable requirements, functioning workflow, justified decisions.	High	High
Project Team	Builder / maintainer	Design, build, test, document, and maintain the workflow.	Clear scope, modular logic, version control, testable requirements, manageable workload.	High	High

2.1 Stakeholder Priorities

- Teacher needs are the primary driver of the functional requirements because teachers are the direct users of the workflow.
- Student needs influence the quality and differentiation of generated materials, but students do not directly operate the core system.
- The Technical Instructor and reviewers need evidence that the workflow satisfies the documented scope and can be tested and explained.
- The project team needs requirements that are clear enough to divide work, build the workflow, and verify completion.

3. User Stories

ID	Actor	User Story	Acceptance Criteria
US-01	Teacher	As a teacher, I want to submit the lesson topic and required instructional information so that the workflow can start from one structured input.	Required fields are accepted and the workflow starts only when mandatory inputs are present.
US-02	Teacher	As a teacher, I want the system to generate a lesson outline so that I can prepare the lesson structure faster.	A clearly labeled lesson outline is produced using the submitted context.
US-03	Teacher	As a teacher, I want student activities generated from the same lesson context so that the activities remain relevant to the lesson.	Activities reflect the topic, grade/level, and learning objective supplied.
US-04	Teacher	As a teacher, I want an assessment and answer key generated together so that I can review questions and correct answers efficiently.	Assessment items and a corresponding answer key are both produced and clearly separated.
US-05	Teacher	As a teacher, I want a differentiated version so that the material can better match the selected learner level or differentiation preference.	The workflow follows the correct branch and produces the requested differentiated output.
US-06	Teacher	As a teacher, I want all generated outputs organized in one clear package so that I can review them before classroom use.	All core outputs are labeled, ordered, and presented for teacher review.
US-07	Teacher	As a teacher, I want a clear message when my input is incomplete or the workflow fails so that I know what to correct or retry.	Invalid input or execution failure is routed to an understandable error/fallback path.
US-08	Technical Instructor / Reviewer	As a reviewer, I want the workflow requirements and behavior documented so that I can verify the project against its intended scope.	Requirements, use cases, workflow behavior, and testable criteria are documented and traceable.
US-09	Project Team	As a project team, we want prompts and workflow logic to be maintainable and versioned so that changes can be tracked during implementation and testing.	Prompt versions and workflow changes are stored and documented in the project repository.

4. Use Cases

ID	Use Case	Actor	Precondition	Main Flow	Outcome
UC-01	Submit instructional input	Teacher	Teacher has access to the input form/trigger.	Teacher submits subject, grade level, topic, learning objective, and relevant options.	Input is received by n8n.
UC-02	Validate input	System / Teacher	UC-01 completed.	Workflow checks required fields and basic validity.	Valid input continues; invalid input goes to clarification/fallback.
UC-03	Generate lesson outline	Teacher / AI workflow	Validated input is available.	AI receives structured context and creates the lesson outline.	Lesson outline output is produced.
UC-04	Generate student activities	Teacher / AI workflow	Shared lesson context is available.	AI generates student activities using the same context.	Activity output is produced.
UC-05	Generate assessment	Teacher / AI workflow	Shared lesson context is available.	AI generates assessment items according to the lesson context.	Assessment output is produced.
UC-06	Generate answer key	Teacher / AI workflow	Assessment output is available.	AI produces a corresponding answer key.	Answer key is produced and associated with the assessment.
UC-07	Generate differentiated version	Teacher / Workflow branch	Differentiation input/condition is available.	Condition routes the workflow to the appropriate differentiation step.	Differentiated version is produced.
UC-08	Compile output for review	Teacher	Core generation steps complete.	Workflow combines and labels all outputs in a consistent structure.	Teacher receives the review-ready package.
UC-09	Handle failure or invalid input	Teacher / System	A validation or execution problem occurs.	Workflow records/identifies the issue and routes to error handling or fallback behavior.	Teacher receives a clear corrective or retry message instead of a misleading final result.

4.1 Main Success Scenario

- Teacher submits valid instructional information.
- n8n trigger receives the data and validates the required fields.
- The workflow passes the same core lesson context through the AI generation stages.
- Lesson outline, activities, assessment, answer key, and differentiated version are generated.
- The outputs are formatted, combined, and presented for teacher review.
- The teacher reviews the material before any classroom use.

4.2 Alternative / Failure Scenario

- If required input is missing, the workflow does not continue to generation and asks for correction.
- If an AI/API step fails or returns unusable output, the workflow uses the defined retry or fallback behavior.
- The workflow should not present an incomplete or misleading package as a successful final result.

5. Functional Requirements

ID	Requirement	Priority
FR-01	The system shall provide a trigger/form to receive teacher instructional input.	Must
FR-02	The system shall accept subject, grade level, topic, and learning objective as core lesson context.	Must
FR-03	The system shall accept lesson duration and differentiation/learner-level preference when relevant.	Should
FR-04	The workflow shall validate mandatory fields before starting AI generation.	Must
FR-05	The workflow shall pass the required shared context between relevant nodes.	Must
FR-06	The workflow shall generate a lesson outline.	Must
FR-07	The workflow shall generate student activities.	Must
FR-08	The workflow shall generate an assessment.	Must
FR-09	The workflow shall generate an answer key corresponding to the assessment.	Must
FR-10	The workflow shall generate a differentiated version using defined conditions/branching.	Must
FR-11	The workflow shall use clear prompt instructions and structured output formats for AI stages.	Must
FR-12	The workflow shall combine and label all core outputs for teacher review.	Must
FR-13	The workflow shall include conditions and branching where the selected input changes workflow behavior.	Must
FR-14	The workflow shall include error handling for failed nodes or AI/API execution problems.	Must
FR-15	The workflow shall provide a fallback/corrective path for invalid input or failed generation.	Must
FR-16	The project shall retain prompts, workflow files/exports, and documentation in the project repository.	Must
FR-17	The teacher shall review generated outputs before classroom use.	Must
FR-18	The workflow may connect to an external API or educational platform after core implementation is stable.	Could

6. Non-Functional Requirements

ID	Category	Requirement	Verification
NFR-01	Usability	The teacher-facing input and output structure should be clear, labeled, and understandable without requiring programming knowledge.	Teacher can identify required inputs and review each generated output section.
NFR-02	Reliability	The workflow should complete successfully for at least 90% of valid-input test runs, aligned with the Task 1 KPI.	Measured from test execution history.
NFR-03	Output Completeness	Every successful core run should produce all required core outputs.	100% of core outputs present in successful runs.
NFR-04	Branch Accuracy	Defined branch/condition test cases should route to the correct workflow path.	100% of defined branch cases during testing.
NFR-05	Error Handling	Defined invalid-input and simulated failure cases should be handled without presenting a misleading final result.	100% of defined error cases during testing.
NFR-06	Content Relevance	Generated outputs should remain relevant to the submitted topic, grade/level, learning objective, and constraints.	At least 85% acceptable results in the agreed review test set.
NFR-07	Performance	Processing time should remain reasonably stable across repeated tests with comparable inputs.	Execution time recorded and compared with the established test baseline.
NFR-08	Security	API keys and credentials must not be hard-coded into shared prompts, documents, or exported public files.	Credentials stored using n8n credential/configuration mechanisms.
NFR-09	Privacy	The core workflow should avoid unnecessary personal student data because Project #9 can operate on lesson-level inputs.	No student personal data required for normal core use.
NFR-10	Maintainability	Prompt templates and workflow logic should be organized, documented, and versioned for team maintenance.	Meaningful filenames/versions and repository updates.
NFR-11	Compatibility	The workflow should run in the team's selected n8n environment and use supported AI/API integrations available to the team.	Verified during environment setup and implementation.

7. Requirements Priority Summary

Priority	Scope
Must Have	Input trigger and validation; five core outputs; data passing; AI transformations; branching/conditions; error handling; fallback path; output formatting; teacher review; documentation.
Should Have	Lesson duration and differentiation controls where relevant; refined prompt structures; clearer user messages and execution logging.
Could Have	External API or educational-platform integration after the core workflow is stable.
Not in Core Scope	Full LMS/student-facing platform, automatic grading of student submissions, or custom AI model training/fine-tuning.

8. Requirements Traceability

User Story	Mapped Functional Requirements	Verification Focus
US-01	FR-01, FR-02, FR-03, FR-04	Input received and validated.
US-02	FR-05, FR-06, FR-11	Lesson outline generated from shared context.
US-03	FR-05, FR-07, FR-11	Student activities generated from shared context.
US-04	FR-08, FR-09, FR-11	Assessment and answer key generated and linked.
US-05	FR-10, FR-13	Correct differentiated branch and output.
US-06	FR-12, FR-17	Complete organized package available for teacher review.
US-07	FR-04, FR-14, FR-15	Invalid/failure case follows clear error or fallback path.
US-08	FR-16	Requirements and workflow evidence documented for review.
US-09	FR-16	Prompts/workflow changes stored and maintainable in repository.

9. Assumptions & Constraints

- The core user is a teacher or teaching professional; students benefit indirectly and do not need direct access in the core project.
- Teacher review is required before AI-generated materials are used.
- The team will implement the core n8n workflow before attempting optional external integration.
- The workflow depends on an available n8n environment and a supported AI model/API configuration.
- No official requirement in the current project scope requires a database, full front-end application, or LMS.
- Requirements may be refined after Technical Instructor feedback, but scope changes should be documented and agreed by the team.